

Phase 2: Visual Analysis with a SIEM Dashboard

Installed and configured Splunk on the Kali machine:

1. Download using this command
`wget -O splunk-9.3.2-d8bb32809498-linux-2.6-amd64.deb`
<https://download.splunk.com/products/splunk/releases/9.3.2/linux/splunk-9.3.2-d8bb32809498-linux-2.6-amd64.deb>
2. Then unpack using:
`sudo dpkg -i splunk-9.3.2-d8bb32809498-linux-2.6-amd64.deb`
3. Start the splunk:
`sudo /opt/splunk/bin/splunk start --accept-license`

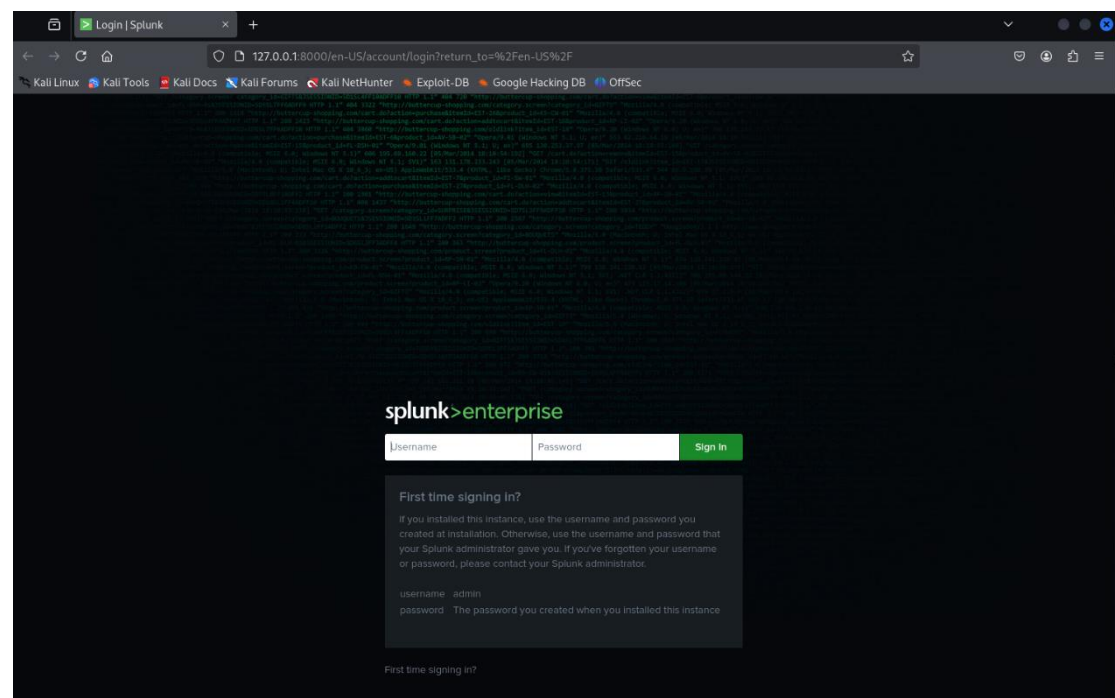
```
(basim@Attacker)-[~/Desktop]
$ sudo /opt/splunk/bin/splunk start --accept-license
sudo: unable to resolve host Attacker: Name or service not known

This appears to be your first time running this version of Splunk.

Splunk software must create an administrator account during startup. Otherwise, you cannot log in.
Create credentials for the administrator account.
Characters do not appear on the screen when you type in credentials.

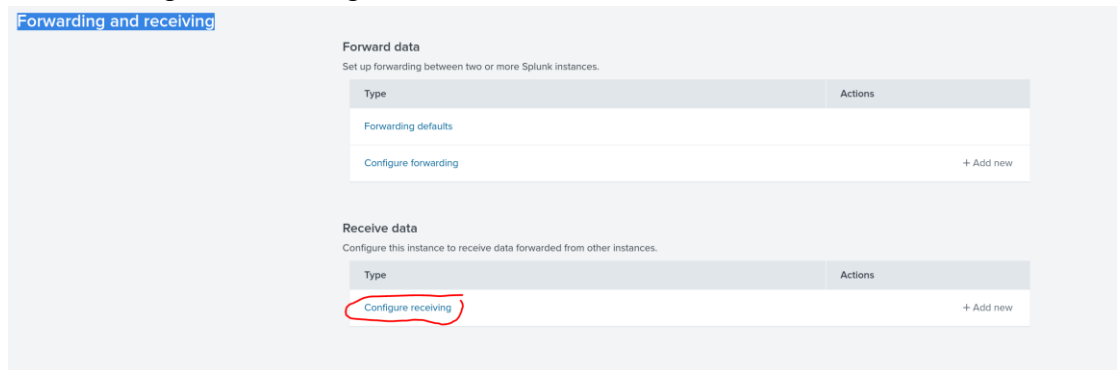
Please enter an administrator username: asd
Password must contain at least:
  * 8 total printable ASCII character(s).
Please enter a new password:
Please confirm new password:
Copying '/opt/splunk/etc/openldap/ldap.conf.default' to '/opt/splunk/etc/openldap/ldap.conf'.
Generating RSA private key, 2048 bit long modulus
```

After registering we can open the web interface:

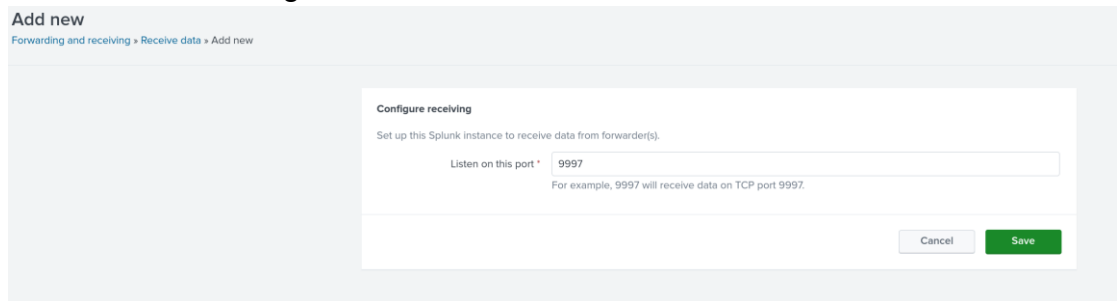


Make the splunk listen to port 9997:

1. Go to Settings → Forwarding and receiving
2. Click Configure receiving



3. Click New Receiving Port and enter 9997



Configured the Splunk Universal Forwarder on Metasploitable3 to send logs to Splunk (port 9997).

ssh to the victim:

```
(basim@Attacker)-[~/Desktop]
$ ssh vagrant@10.0.2.15
vagrant@10.0.2.15's password:
Welcome to Ubuntu 14.04.6 LTS (GNU/Linux 3.13.0-170-generic x86_64)

 * Documentation:  https://help.ubuntu.com/
New release '16.04.7 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Thu May  1 16:07:31 2025
vagrant@metasploitable3-ub1404:~$
```

Install Splunk Universal Forwarder:

```
vagrant@metasploitable3-ub1404:~$ wget -O splunkforwarder-9.4.1-amd64.deb "https://download.splunk.com/products/universalforwarder/releases/9.4.1/linux/splunkforwarder-9.4.1-e3bdab203ac8-linux-amd64.deb"
--2025-05-02 16:31:01-- https://download.splunk.com/products/universalforwarder/releases/9.4.1/linux/splunkforwarder-9.4.1-e3bdab203ac8-linux-amd64.deb
Resolving download.splunk.com (download.splunk.com)... 108.159.236.91, 108.159.236.108, 108.159.236.116, ...
Connecting to download.splunk.com (download.splunk.com)|108.159.236.91|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 99829222 (94M) [binary/octet-stream]
Saving to: 'splunkforwarder-9.4.1-amd64.deb'

100%[<img alt="Ubuntu logo" data-bbox="350 400 650 600"/>] 99,829,222  1.44MB/s  in 71s

2025-05-02 14:32:13 (1.33 MB/s) - 'splunkforwarder-9.4.1-amd64.deb' saved [99829222/99829222]

vagrant@metasploitable3-ub1404:~$ ls
splunkforwarder-9.4.1-amd64.deb  splunkforwarder-9.4.1-amd64.deb  VBoxGuestAdditions.iso
vagrant@metasploitable3-ub1404:~$ sudo dpkg -i splunkforwarder-9.4.1-amd64.deb
Selecting previously unselected package splunkforwarder.
(Reading database ... 127930 files and directories currently installed.)
Preparing to unpack splunkforwarder-9.4.1-amd64.deb ...
no need to run the pre-install check
Unpacking splunkforwarder (9.4.1) ...
Setting up splunkforwarder (9.4.1) ...
find: /opt/splunkforwarder/lib/python3.7/site-packages: No such file or directory
find: /opt/splunkforwarder/lib/python3.9/site-packages: No such file or directory
complete
vagrant@metasploitable3-ub1404:~$
```

Run it:

```
vagrant@metasploitable3-ub1404:~$ sudo /opt/splunkforwarder/bin/splunk start --accept-license
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"

This appears to be your first time running this version of Splunk.

Splunk software must create an administrator account during startup. Otherwise, you cannot log in.
Create credentials for the administrator account.
Characters do not appear on the screen when you type in credentials.

Please enter an administrator username: victim
Password must contain at least:
 * 8 total printable ASCII character(s).
Please enter a new password:
Please confirm new password:
Creating unit file...
Error calling execve(): No such file or directory
Error launching command: No such file or directory
Failed to create the unit file. Please do it manually later.

Splunk> See your world. Maybe wish you hadn't.

Checking prerequisites...
  Checking mgmt port [8089]: open
    Creating: /opt/splunkforwarder/var/lib/splunk
    Creating: /opt/splunkforwarder/var/run/splunk
    Creating: /opt/splunkforwarder/var/run/splunk/appserver/i18n
    Creating: /opt/splunkforwarder/var/run/splunk/appserver/modules/static/css
    Creating: /opt/splunkforwarder/var/run/splunk/upload
    Creating: /opt/splunkforwarder/var/run/splunk/search_telemetry
    Creating: /opt/splunkforwarder/var/run/splunk/search_log
    Creating: /opt/splunkforwarder/var/spool/splunk
    Creating: /opt/splunkforwarder/var/spool/dironcache
    Creating: /opt/splunkforwarder/var/lib/splunk/authdb
    Creating: /opt/splunkforwarder/var/lib/splunk/hashdb
    Creating: /opt/splunkforwarder/var/run/splunk/collect
    Creating: /opt/splunkforwarder/var/run/splunk/sessions
New certs have been generated in '/opt/splunkforwarder/etc/auth'.
  Checking conf files for problems...
    Done
  Checking default conf files for edits...
  Validating installed files against hashes from '/opt/splunkforwarder/splunkforwarder-9.4.1-e3bdab203ac8-linux-amd64-manifest'
    All installed files intact.
    Done
All preliminary checks passed.

Starting splunk server daemon (splunkd)...
PYTHONHTTPSVERIFY is set to 0 in splunk-launch.conf disabling certificate validation for the httplib and urllib libraries shipped with the embedded Python interpreter; must be used security
```

Monitored and uploaded logs like /var/log/auth.log.:

```
vagrant@metasploitable3-ub1404:~$ sudo /opt/splunkforwarder/bin/splunk add monitor /var/log/auth.log -index main -sourcetype linux_secure
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
Splunk username: victim
Password:
Added monitor of '/var/log/auth.log'.
vagrant@metasploitable3-ub1404:~$ sudo /opt/splunkforwarder/bin/splunk add monitor /var/log/syslog -index main -sourcetype linux_secure
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
Added monitor of '/var/log/syslog'.
vagrant@metasploitable3-ub1404:~$
```

Forward it to our server:

```
vagrant@metasploitable3-ub1404:~$ sudo /opt/splunkforwarder/bin/splunk add forward-server 10.0.2.5:9997
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
Added forwarding to: 10.0.2.5:9997.
vagrant@metasploitable3-ub1404:~$
```

Run Splunk searches to detect failed and successful login attempts. (I will provide the csv files for the attempts in github)

splunk>enterpriseApps

SearchAnalyticsDatasetsReportsAlertsDashboards

New Search

index=* sourcetype=linux_secure "failed password"

6,304 events (5/1/25 5:00:00.000 PM to 5/2/25 5:45:44.000 PM)No Event Sampling

Events (6,304)PatternsStatisticsVisualization

Format TimelineZoom OutZoom to SelectionDeselect

< Hide FieldsAll Fields

SELECTED FIELDS
host 1
source 1
sourcetype 1

INTERESTING FIELDS
date_hour 3
date_minute 28
date_month 1
date_second 60
date_wday 2
date_year 1
date_zone 1
index 1
linecount 1
pid 1001
process 1
punct 3
splunk_server 1
timeendpos 1
timelastpos 1

Extract New Fields

ListFormat20 Per Page

i	Time	Event
>	5/2/25 5:26:03.000 PM	metasploit3-ub1404 sshd[2710]: failed password for invalid user basin from 10.0.2.5 port 50296 ssh2 host = metasploit3-ub1404 source = /var/log/auth.log sourcetype = linux_secure
>	5/2/25 4:54:14.000 PM	metasploit3-ub1404 sshd[2424]: failed password for vagrant from 10.0.2.5 port 51116 ssh2 host = metasploit3-ub1404 source = /var/log/auth.log sourcetype = linux_secure
>	5/2/25 4:54:12.000 PM	metasploit3-ub1404 sshd[2422]: failed password for vagrant from 10.0.2.5 port 51110 ssh2 host = metasploit3-ub1404 source = /var/log/auth.log sourcetype = linux_secure
>	5/2/25 4:54:10.000 PM	metasploit3-ub1404 sshd[2420]: failed password for vagrant from 10.0.2.5 port 51104 ssh2 host = metasploit3-ub1404 source = /var/log/auth.log sourcetype = linux_secure
>	5/2/25 4:54:09.000 PM	metasploit3-ub1404 sshd[2418]: failed password for vagrant from 10.0.2.5 port 51102 ssh2 host = metasploit3-ub1404 source = /var/log/auth.log sourcetype = linux_secure
>	5/2/25 4:54:07.000 PM	metasploit3-ub1404 sshd[2416]: failed password for vagrant from 10.0.2.5 port 52110 ssh2 host = metasploit3-ub1404 source = /var/log/auth.log sourcetype = linux_secure
>	5/2/25 4:54:05.000 PM	metasploit3-ub1404 sshd[2414]: failed password for vagrant from 10.0.2.5 port 52094 ssh2 host = metasploit3-ub1404 source = /var/log/auth.log sourcetype = linux_secure
>	5/2/25 4:54:02.000 PM	metasploit3-ub1404 sshd[2412]: failed password for vagrant from 10.0.2.5 port 52092 ssh2 host = metasploit3-ub1404 source = /var/log/auth.log sourcetype = linux_secure
>	5/2/25 4:54:00.000 PM	metasploit3-ub1404 sshd[2409]: failed password for vagrant from 10.0.2.5 port 52082 ssh2 host = metasploit3-ub1404 source = /var/log/auth.log sourcetype = linux_secure
>	5/2/25 4:53:58.000 PM	metasploit3-ub1404 sshd[2407]: failed password for vagrant from 10.0.2.5 port 50162 ssh2 host = metasploit3-ub1404 source = /var/log/auth.log sourcetype = linux_secure
>	5/2/25 4:53:57.000 PM	metasploit3-ub1404 sshd[2405]: failed password for vagrant from 10.0.2.5 port 50150 ssh2 host = metasploit3-ub1404 source = /var/log/auth.log sourcetype = linux_secure
>	5/2/25 4:53:55.000 PM	metasploit3-ub1404 sshd[2403]: failed password for vagrant from 10.0.2.5 port 50144 ssh2 host = metasploit3-ub1404 source = /var/log/auth.log sourcetype = linux_secure
>	5/2/25 4:53:52.000 PM	metasploit3-ub1404 sshd[2401]: failed password for vagrant from 10.0.2.5 port 50142 ssh2 host = metasploit3-ub1404 source = /var/log/auth.log sourcetype = linux_secure
>	5/2/25 4:53:50.000 PM	metasploit3-ub1404 sshd[2399]: failed password for vagrant from 10.0.2.5 port 50132 ssh2 host = metasploit3-ub1404 source = /var/log/auth.log sourcetype = linux_secure
>	5/2/25 4:53:48.000 PM	metasploit3-ub1404 sshd[2397]: failed password for vagrant from 10.0.2.5 port 50594 ssh2 host = metasploit3-ub1404 source = /var/log/auth.log sourcetype = linux_secure
>	5/2/25 4:53:47.000 PM	metasploit3-ub1404 sshd[2395]: failed password for vagrant from 10.0.2.5 port 50590 ssh2 host = metasploit3-ub1404 source = /var/log/auth.log sourcetype = linux_secure
>	5/2/25 4:53:44.000 PM	metasploit3-ub1404 sshd[2391]: failed password for vagrant from 10.0.2.5 port 50498 ssh2 host = metasploit3-ub1404 source = /var/log/auth.log sourcetype = linux_secure
>	5/2/25 4:53:42.000 PM	metasploit3-ub1404 sshd[2349]: failed password for vagrant from 10.0.2.5 port 50484 ssh2 host = metasploit3-ub1404 source = /var/log/auth.log sourcetype = linux_secure
>	5/2/25 4:53:40.000 PM	metasploit3-ub1404 sshd[2347]: failed password for vagrant from 10.0.2.5 port 50482 ssh2 host = metasploit3-ub1404 source = /var/log/auth.log sourcetype = linux_secure

New Search

index=* sourcetype=linux_secure "Accepted password"

5 events (5/1/25 6:00:00.000 PM to 5/2/25 6:28:21.000 PM)No Event Sampling

Events (5)PatternsStatisticsVisualization

Format TimelineZoom OutZoom to SelectionDeselect

< Hide FieldsAll Fields

SELECTED FIELDS
host 1
source 1
sourcetype 1

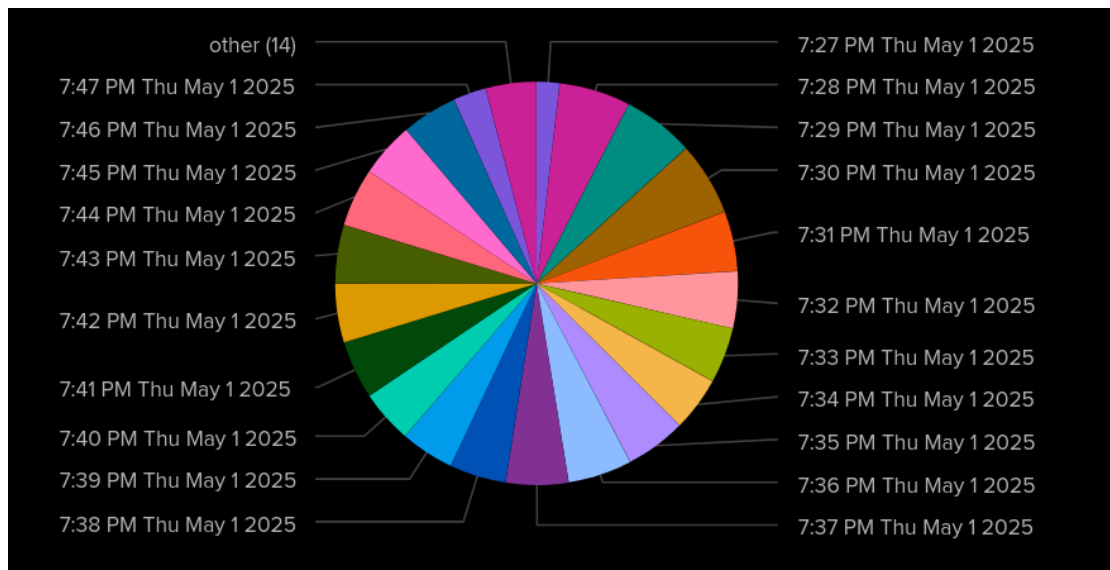
INTERESTING FIELDS
date_hour 4
date_minute 5
date_month 1
date_second 5
date_wday 2
date_year 1
date_zone 1
index 1
linecount 1
pid 5

Extract New Fields

ListFormat20 Per Page

i	Time	Event
>	5/2/25 6:27:15.000 PM	metasploit3-ub1404 sshd[3081]: Accepted password for vagrant from 10.0.2.5 port 60572 ssh2 host = metasploit3-ub1404 source = /var/log/auth.log sourcetype = linux_secure
>	5/2/25 5:59:50.000 PM	metasploit3-ub1404 sshd[3482]: Accepted password for vagrant from 10.0.2.5 port 56746 ssh2 host = metasploit3-ub1404 source = /var/log/auth.log sourcetype = linux_secure
>	5/2/25 5:26:24.000 PM	metasploit3-ub1404 sshd[2754]: Accepted password for vagrant from 10.0.2.5 port 48634 ssh2 host = metasploit3-ub1404 source = /var/log/auth.log sourcetype = linux_secure
>	5/2/25 4:54:14.000 PM	metasploit3-ub1404 sshd[2426]: Accepted password for vagrant from 10.0.2.5 port 51130 ssh2 host = metasploit3-ub1404 source = /var/log/auth.log sourcetype = linux_secure
>	5/1/25 7:47:32.000 PM	metasploit3-ub1404 sshd[4666]: Accepted password for vagrant from 10.0.2.5 port 60556 ssh2 host = metasploit3-ub1404 source = /var/log/auth.log sourcetype = linux_secure

Chart of the attack pattern:



Conclusion:

Splunk successfully detected and visualized Hydra brute-force attempts on the victim machine. The data showed a high frequency of failed SSH logins from a single IP (the attacker).